PORTFOLIO / INDEPENDENT STUDY

Autonomous Mobility Business Feasibility

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PORTFOLIO

Autonomous Mobility Business Execution Architecture

— Driving autonomous mobility business development through AD/ADAS system foundation —

Autonomous Mobility Business Definition

This architecture is defined as a control system where KPI, system behavior, and business outcomes are directly connected. Each KPI is traced to the system behavior that moves it and to its effect on revenue, cost, and return.
The objective is to build an AI-enabled autonomous mobility business architecture that improves utilization, load factor, and unit economics.
This is achieved by integrating mobility, logistics, energy, and urban services through a common operating architecture, while structurally improving:

- utilization
- load factor
- unit economics

This proposal defines a business architecture that makes this system executable, economically grounded, and deployable.

Autonomous Mobility Business Execution

This proposal defines a phased development and deployment plan for autonomous mobility business execution.
Today's mobility systems fail to scale because demand, fleet, energy, and operations are still managed separately.
This architecture integrates these elements into a unified autonomous mobility business structure, using autonomy as the foundation and AI as the means to operate autonomous mobility at business scale. System performance directly translates into revenue through utilization, load factor, and energy efficiency:

• Demand creation synchronized with dispatch
• Utilization maximization through modular vehicle operation
• Load factor optimization through dynamic vehicle composition
• Energy cost reduction through on-site hydrogen production
• Real-time operational control through distributed node coordination and AI-enabled optimization

Autonomous Mobility Business Architecture

Objective: To define an executable autonomous mobility business structure where AI-driven control logic is applied through autonomous driving to integrate demand, operations, vehicles, and infrastructure.
This architecture shifts mobility from fragmented services into a unified operating system where utilization, load factor, and energy efficiency are structurally optimized.
AI is applied as controlled logic across four layers.

1. Demand Orchestration

2030: Coordinated AI Assistant
A calendar-based interface connects user schedules with service availability, enabling demand generation through matching and reservation-linked mobility.
2035: Trusted AI Partner
A 4D digital platform integrated with map infrastructure synchronizes city-state and user behavior in real time, enabling continuous demand generation and capture to maximize load factor.

2. Operation Control

2030: Integrated Control
Energy management and dispatch are coordinated through decision logic that can later be augmented by AI, enabling stable and efficient operations under constrained conditions.
2035: Synchronized Control
Real-time synchronization of in-car and out-car hydrogen and electricity flows creates the control foundation for AI-assisted optimization, while structurally improving business viability.

3. Fleet Vehicle

2030: Swap Carrier
A modular vehicle structure enables continuous operation through container exchange, maximizing utilization.
2035: Composed Carrier
Vehicles dynamically couple and reconfigure during operation, enabling real-time redistribution of passengers and cargo to maximize load factor.

4. Distributed Infrastructure

2030: Urban Node
Distributed hubs provide localized energy and compute support, enabling operational independence and cost efficiency.
2035: Node Mesh
Distributed infrastructure is synchronized as a network, enabling real-time coordination between vehicles, energy, and computation.

This architecture enables a system where mobility, energy, computing, and AI-based decision-making operate as a unified layer, enabling scalable autonomous mobility business operation.

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AI-Enabled Urban Mobility Roadmap

Autonomous mobility becomes deployable only when a production-ready driving stack, closed-loop learning, auditable safety, fleet operations and service economics are developed as one system. This roadmap first establishes that industrial baseline, then extends AI control from vehicles to distributed infrastructure and city-scale operations.

Industry Baseline & Capability Build-Up

Capability	Industry Baseline	Roadmap Focus
Product / Release Governance	Common requirements, configuration control, release evidence and change management across vehicles, sites and ODDs	Stage deployment through safety, service-readiness, availability, recovery-time and unit-economics gates.
Driver / Sensor Platform	Multimodal camera–lidar–radar perception, overlapping coverage, redundancy and deployment across multiple vehicle classes.	Common driving stack, hardware abstraction, calibration, fail-operational design and rapid vehicle onboarding.
Data / Active Learning	Fleet-scale data collection, anonymization, annotation, rare-event mining and uncertainty-based active learning.	Closed loop: detect → anonymize → select → label → train → validate → staged release.
Simulation / V&V	Real-event replay, scenario variation, closed-course and public-road tests, synthetic simulation and automated regression.	ODD-specific scenario coverage, acceptance criteria, critical-event recall, safety evidence and validation gates.
Safety / Remote Support	ODD safety cases, independent evidence, fallback/MRM, remote monitoring and contextual assistance with clear authority boundaries.	Claims, evidence, escalation, incident response, cybersecurity, rollback, audit trail and remote-support limits.
Fleet / Service Operations	Integrated dispatch, maintenance, charging, incident management, customer support and feedback from daily autonomous operations.	Connect the driving stack with fleet and service control; manage SLA, availability, intervention, energy and cost/hour.
Scale / Economics	Reusable software/hardware platforms, lower-cost sensors and compute, standardized integration and multi-vehicle deployment.	Track utilization, launch lead time, ODD rollout speed and lifecycle revenue; separate common and use-case-specific functions.

The industrial baseline combines a production-ready driver, closed-loop learning, scenario-based V&V, auditable safety, remote support, fleet operations and repeatable ODD deployment. The roadmap establishes these capabilities first, then differentiates through city data, infrastructure coordination, demand orchestration and business optimization.

AI-Controlled Mobility System

This section defines how AI functions as control logic across layers and is executed through autonomous driving systems. AI is applied differently at each layer — from demand prediction to system-wide coordination — but all functions are integrated through a unified control structure that enables real-time execution at city scale.

Layer	Cloud	Distributed Infrastructure	Fleet Vehicle
Perception	2030: Env-state estimation (fusion, uncertainty propagation) / 2035: City-wide world-state estimation (global sync, uncertainty minimization)	2030: Roadside perception augmentation (edge fusion, low-latency inference) / 2035: Cooperative perception (shared state, spatiotemporal fusion)	2030: Onboard perception (LiDAR / Camera / Radar, ≤10ms) / 2035: Infra-coordinated perception (shared scene state)
Planning	lispatch optimization (forecast, :onstrained search) / 2035: City nission optimization (global :cheduling, real-time recompute)	2030: Zone operation optimization (routing / berth / node) / 2035: Real-time replanning (node sync, global constraints)	2030: Local path / motion planning / 2035: Cooperative trajectory planning
Control	2030: Learning-based control (constraints, stability) / 2035: Global closed-loop optimization (adaptive)	2030: Edge feedback control (low-latency, constraints) / 2035: Distributed synchronized control (coordination, load)	2030: Motion control (MPC, ≤50ms, constraint-aware) / 2035: High-frequency adaptive control (≤10ms, cooperative)
Coordination	2030: Fleet-state coordination (dispatch-state sync) / 2035: Fleet-wide coordination (global sync, decision integration)	2030: Node-state coordination (local sync) / 2035: Node Mesh synchronization	2030: Vehicle-level coordination / 2035: Platoon coordination

ODD / DDT as Revenue Driver

ODD / DDT transition defines idle time and directly determines utilization and revenue structure.

DDT Loss = (1-TSR) x Recovery time + TOT, Revenue = OAR x UAC x(1÷(Trip+(1-TSR) x Recovery + TOT) x FLC x Load

Execution Sequence & Exit Gates

The roadmap advances only when technical evidence, operational readiness and unit economics are verified together.

Phase	Capability Build	Exit Gate	Business Outcome
1. Industrialize	Common driving and sensor stack; closed-loop data; scenario V&V; auditable safety case; remote operations and field support.	Repeatable launch within the initial ODD, with controlled fallback, rollback, incident response and configuration management.	Production service with measurable safety, availability, intervention rate, recovery time and cost/hour.
2. Federate	Fleet orchestration; RSU / OBU coordination; shared data and simulation; distributed infrastructure and service control.	Cross-vehicle and cross-site operation under common interfaces, authority boundaries, release procedures and KPI governance.	Lower intervention, recovery time and idle capacity; faster vehicle, site and ODD onboarding.
3. Scale	4D demand orchestration; Node Mesh; energy-compute integration; evidence-led quantum computing, sensing and optimization.	Multi-ODD, multi-service and multi-city performance with resilience, traceability and viable unit economics.	Scalable urban mobility platform with recurring service, infrastructure and lifecycle revenue.

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Urban AI Operating Architecture

An integrated capability progression from real-world understanding and coordinated safety to governed learning and city-scale operations.

1. UNDERSTAND & PROTECT Real-world intelligence and coordinated safety	2. CONNECT & IMPROVE Governed data, validation and continuous learning	3. OPERATE & SCALE Distributed execution and city-scale orchestration
AI Vision Engine — Builds a multimodal understanding of people, vehicles, infrastructure, road conditions and the surrounding urban environment.	Infra Hub — Connects roadside, mobility, building and public-service infrastructure through secure, governed and interoperable interfaces.	4D demand orchestration — Predicts where, when and why demand will emerge across mobility, logistics, energy and public services.
Behavior AI — Predicts human and vehicle intent, interaction patterns, anomalies and uncertainty to support context-aware decisions.	Data Fabric — Federates vehicle, city, infrastructure and operational data without forcing every participant into a single centralized system.	Rare-event learning — Prioritizes edge cases, near misses and low-frequency, high-impact events for targeted learning and validation.
Drive Sync Assist — Synchronizes driving support, shared awareness and timely intervention across the driver, vehicle and surrounding environment.	Digital KAIZEN — Converts operational evidence into simulation, validation, controlled deployment and measurable continuous improvement.	RSU / OBU coordination — Distributes perception, routing, safety and operational tasks across roadside and onboard systems.
Integrated ANZEN — Integrates detection, prediction, risk assessment and intervention, including graduated assistance, fallback and minimum-risk maneuvers.	Scenario simulation and validation — Tests policies, ODD changes, service patterns, traffic conditions and rare events before field deployment.	Node Mesh synchronization — Coordinates vehicles, roadside nodes and local service hubs as a resilient, distributed operating network.
Multimodal real-world understanding — Fuses cameras, sensors, maps, V2X signals and contextual data into a consistent operational picture.	Operational feedback loop — Captures outcomes, deviations, user behavior and system performance to update models, rules and operating procedures.	Energy-compute integration — Allocates charging, storage, edge computing and fleet tasks against demand, cost, capacity and resilience.
Predictive risk and uncertainty management — Identifies emerging conflicts early enough to preserve human decision time and maintain a safe operating margin.	MBD / digital-twin lifecycle loop — Links design assumptions, simulation results, field evidence and software or policy updates throughout the lifecycle.	Quantum computing and sensing — Explores optimization and sensing gains for high-dimensional mobility, infrastructure and urban-service operations.
Coordinated safety — Aligns human handover, vehicle fallback, infrastructure assistance and accountable decision-making across the operating environment.	Governed continuous improvement — Releases only validated updates with traceability, approval, rollback capability and measurable safety evidence.	Quantum-enhanced perception — Applies advanced sensing and inference only where they demonstrate meaningful gains over classical approaches.
Shared scene awareness — Makes relevant hazards, intent and operating context understandable across people, vehicles, infrastructure and service operators.	City-data integration — Connects mobility, energy, logistics and public-service operations through governed, reusable data products.	Full-city synchronization — Coordinates cross-domain actions while preserving local autonomy, safety boundaries and service continuity.
Safety-to-ODD expansion — Extends system capability only after evidence demonstrates acceptable risk and operational readiness in each target condition.	Closed-loop execution — Deploys approved models, policies and operating rules back to vehicles, RSUs, infrastructure and service operations.	Autonomous city-state optimization — Continuously balances safety, service quality, capacity, energy consumption, resilience and accessibility.
Human trust and adoption — Explains system state, preserves appropriate human authority and captures acceptance signals from users and operators.	Governance, KPI and business control — Monitors safety, availability, unit economics, energy, inclusion, service quality and lifecycle performance.	Scalable business execution — Converts technical gains into repeatable operations, partnerships, asset management and sustainable lifecycle value.

Cross-Layer Decision Rights & Release Gates

Control Domain	Primary Authority	Evidence Returned Upward	Release / Deployment Gate
Vehicle / Edge	Immediate perception, motion control, fallback, minimum-risk manoeuvre and local safety intervention.	Uncertainty, hazard traces, interventions, sensor and actuator health, and safe-state results.	Latency, hazard-response coverage, fail-operational capability and safe-state completion.
Fleet / Infrastructure	Dispatch, routing, RSU / OBU coordination, remote assistance, node control and energy-compute allocation.	Availability, dwell and recovery time, node congestion, service exceptions, energy use and capacity.	SLA, intervention rate, recovery time, operational continuity and ODD-support readiness.
Cloud / City	Model and policy release, scenario validation, governance, cross-domain learning and city-scale optimization.	Scenario coverage, causal evidence, audit trail, service quality, cost and lifecycle-performance KPIs.	Traceability, rollback readiness, validated KPI improvement, resilience and sustainable unit economics.

Validated policies, models and operating rules move downward; evidence, uncertainty, incidents and business KPIs move upward. Safety authority remains local, while learning and optimization remain globally governed, measurable and auditable.

Across these three layers, the operating cycle remains Observe → Understand → Learn → Evaluate → Execute → Observe Again. Real-world signals are interpreted and protected at the edge; governed data is federated and converted into validated improvements; approved policies and models are then deployed back to vehicles, RSUs, infrastructure and urban operations. Evidence and learning move upward, while only validated policies and control updates move downward. The architecture scales from coordinated assistance to cross-domain orchestration across mobility, logistics, energy and public services, while preserving human accountability, fallback and rollback. Quantum computing, sensing and quantum-enhanced perception are introduced as evidence-led accelerators—not assumptions—and must demonstrate measurable gains in safety, resilience, service quality, capacity or lifecycle economics before deployment. The result is a practical path from isolated AI functions to a trusted, continuously improving urban operating system.

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Layer-integrated Control & Business Architecture

This architecture defines how business requirements are translated into executable system design through AI-driven control and autonomous driving.
It connects KPI targets with technical execution, ensuring that system performance directly drives economic outcomes.

Executable mobility operation

This section maps business KPIs to control logic and AI-relevant technical conditions required to achieve them.

	2030			2035		
	Logic	**Input**	**Target**	**Logic**	**Input**	**Target**
Perception	Match ≤500ms [A] Update ≤5min Trigger ≤1s Conv ≤1min Err ≤20% Succ ≥85%	Cal ±5min Res ≤1s Avail ≤500ms	CVR ≥23% Wait ≤6min Cancel ≤12% Eff conv≥19.6% Match ≤500ms Err ≤20% Trigger ≤1s	Match ≤100ms [A] Update ≤1s Trig ≤100ms Conv ≤10s Err ≤5% Reflect ≥95%	Behavior ±1s History ≤10s Avail ≤100ms	CVR ≥35% Wait ≤3min Cancel ≤5% Eff ≥33.3% Match ≤100ms Err ≤5% Trig ≤100ms
Planning	Disp ≤1min [B] Route ≤5s Alloc ≤10min Loop ≤5min Succ ≥95%	Fleet ±50m ETA err ≤3min Traffic ≤30s	Trips ≥73/d Loaded ≥14/d Resp ≤8min ETA err ≤3min Empty ≤25% Util ≥75% Route ≤5s	Disp ≤10s [B] Route ≤1s Alloc ≤1min Loop ≤1min E-int ≥95% Succ ≥99%	Fleet ±5m Traffic ±1s Sync ≤5s	Trips ≥140/d Load ≥112/d Resp ≤5min ETA err ≤2min Empty ≤15% Util ≥85% Route ≤1s
Control	MPC ≤50ms [C] Traj ≤50ms Conv ≤100ms Stability ±5% Fail-safe ≤100ms	Percep ≥95% Plan ≤100ms V2X ≤20ms	Price ≥¥2,200 Ctrl ≤20ms Plan ≤100ms Err ≤30cm Intv ≤1/100km Avail ≥87.5%	MPC ≤10ms [C] Traj ≤10ms Conv ≤10ms Stability ±5% Coop ≥95% Fail-safe ≤10ms	Percep ≥99.9% Plan ≤50ms Sync ≤10ms	Price ≥¥2,000 Ctrl ≤10ms Plan ≤50ms Err ≤5cm Intv ≤1/10,000km Avail ≥95.8%
Coordination	Alloc ≤1min Compute ≤1s Refuel ≤5min Conv ≤10min S-D ≥0.78 Load err ≤10%	Energy ±5% [D] Comp ≤100ms Sync ≤1min	OPEX bound Comp ≤150ms Eff ≥75% Node util ≥78% Eff energy ≥58.5% Refuel ≤2/d Var ≤±5%	Alloc ≤100ms Comp ≤10ms Refuel ≤1min Strategy ≤1h Conv ≤1min S-D ≥0.96 Load err ≤3%	Energy ±1% [D] Comp ≤10ms Sync ≤10s	OPEX≈¥202.8K/d Comp ≤50ms Eff ≥90% Node util ≥96% Eff energy ≥86.4% Refuel ≤1/d Var ≤±1%

Operating index (dimensionless) = Utilization × Load Factor × Energy Efficiency; profit still requires price, volume, and cost.
Available service time = scheduled time − [(1−TSR)×recovery time + TOT]; Revenue = Trips × Seats × Load × Fare.

Layer-Integrated Business Architecture

This section defines business requirements and constraints. It translates them into executable system architecture.
AI is applied as controlled logic across four layers.

	2030		2035	
	Technology	**KPI**	**Technology**	**KPI**
Service Orchestration	・Calendar-driven Demand Orchestration Engine ・Service Reservation ・Availability Matching Engine ・Reservation-triggered Dispatch ・Linkage API ・Customer Touchpoint PaaS ・Context-aware Notification	CVR ≥23% Wait ≤6min Cancel ≤12% Match ≤500ms [A] Avail ≤500ms Res ≤1s Cal ±5min Trigger ≤1s	・Behavioral AI Orchestration ・Agent-Demand Control ・4D Digital Twin ・Travel-time Monetization ・City Service Optimization ・Fleet Camera social sensing ・IFTTT Trigger Dispatch ・Autonomous Intent Agent	CVR ≥35% Wait ≤3min Cancel ≤5% Match≤100ms [A] Avail ≤100ms Behavior ±1s History ≤10s Reflect ≥95%
Operation Control	・Dispatch Optimization (VRP) ・Real-time Routing ・Closed-loop Dispatch ・Demand Prediction ・Energy-aware Routing ・Demand-Supply Balancing ・ETA Prediction ・Capacity Allocation ・ETA Feedback Control ・Queue Optimization	Trips ≥73/d [B] Load ≥14/d Resp ≤8min ETA err ≤3min Route ≤5s Alloc ≤10min Fleet ±50m Traffic ≤30s Empty ≤25% Util ≥75%	・Integrated TMS ・Fleet-cooperative Control ・Energy-Transport Optimizer ・Quantum-inspired Optimization ・In / Out Energy Sync ・Real-time Replanning ・Constraint-aware Optimization ・Multi-agent Coordination ・Global State Synchronization ・Predictive Dispatch	Trips ≥140/d [B] Load ≥112/d Resp ≤5min ETA err ≤2min Route ≤1s Alloc ≤1min Fleet ±5m Traffic ±1s Empty ≤15% Util ≥85%
Fleet Vehicle	・Swappable Vehicle Module ・MS-SOFC＋SSB ・LH2 Tank TypeⅣ Subframe ・MgH2 MH Main chassis ・PMSM Distributed Drive ・SiC Inverter ・SbW (Dual Motor Redundant) ・BbW (Electro-Hydraulic +Regen) ・CFRP Leaf + Active Suspension ・CFRP Spoke + CTIS Tire	Avail ≥87.5% [C] Err ≤30cm MPC ≤50ms Traj ≤50ms Conv ≤100ms Stability ±5% Fail-safe ≤100ms Percep ≥95% Plan ≤100ms Intv ≤1/100km	・Dynamic Vehicle Coupling ・Onboard Mill+ Al Air Bat ・HTS coil winded LH2 Tank ・H- Conductor adapted MH ・Real-time energy optimization ・Thermal-compute coupling ・High-frequency power control ・Quantum-enhanced LiDAR ・Quantum magnetometer ・Atomic clock	Avail ≥95.8% [C] Err ≤5cm MPC ≤10ms Traj ≤10ms Conv ≤10ms Stability ±5% Coop ≥95% Fail-safe ≤10ms Percep ≥99.9% Plan ≤50ms Intv ≤1/10,000km
Distributed Infrastructure	・High-precision docking ・Wind-driven Mill ・Wind-driven AMR ・SOFC + Ammonia Tank ・Superconducting Power Cable ・6G Communication Network ・Standardized energy interface ・Low-latency communication ・Sub-ms time synchronization	Eff ≥75% Node util ≥78% Energy ±5% [D] Compute ≤1s Sync ≤1min Refuel ≤5min S-D ≥0.78 Load err ≤10% Var ≤±5%	・Precision docking ・Ultra-low latency V2X sync ・Water-pipeline Mill ・Pipeline-integrated AMR ・PCFC ・H-conductive BIMH Storage ・HTS-coiled LH2 Pipeline ・Quantum Communication ・Energy-transport interface	Eff ≥90% Node util ≥96% Energy ±1% [D] Compute ≤10ms Sync ≤10s Refuel ≤1min S-D ≥0.96 Load err ≤3% Var ≤±1%

I design systems where business requirements and technical execution are structurally aligned.

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System Design for Mobility Feasibility

This page defines scenario design targets for Energy Efficiency = 0.58 (=58%), Utilization = 0.75 (=18h/day ÷ 24h/day), and Load Factor = 0.70 (=2.8pax ÷ 4pax). They are not proven outcomes; each target requires metered validation through integrated system architecture.

The design is structured across energy supply [kWh/day], vehicle dynamics and response [ms, cm, Hz], and control and computation [ms, Hz]. Net traction demand is 2,080.5kWh/day (=28.5kWh/trip × 73trips/node). At 60% electrical efficiency, required fuel-LHV input is 3,467.5kWh/day and the hydrogen-equivalent requirement is 104.1kg/day (=3,467.5 ÷ 33.3). Annual PV energy can exceed annual net demand, but an annual ratio does not prove 24-hour supply without power, storage, loss, and weather modelling.

Vehicle and control design ensure "motion sickness free" operation, defined by control latency ≤20ms, decision latency ≤100ms, positioning error ≤10cm, and vibration ≤0.3Hz. These are maintained through superconductivity enabled by liquid hydrogen cooling and real-time control via HPC operating at 100Hz (=1 ÷ 0.01sec). The HPC value is a design target and does not by itself prove continuous satisfaction or ODD expansion.

Block	Variable	Corrected value	Formula / definition	KPI link
KPI	Energy Efficiency	0.58 target	Σ useful traction output ÷ Σ purchased energy input	Energy
	Utilization	0.75 (=18÷24)	18h/day ÷ 24h/day	Operation
	Load Factor	0.70 (=2.8÷4)	2.8pax ÷ 4pax	Demand
Constraint	Operation Hours	24h/day	operating-window target	Utilization
	Stability	0.95	1 − 0.05	Energy
	Throughput	40veh/h	60min ÷ 1.5min	Infra
Energy	Net traction demand	2,080.5kWh/day	28.5×73	Energy
	Fuel-LHV input	3,467.5kWh/day	2,080.5÷0.60	Energy
	SOFC electrical efficiency	0.60	net electricity ÷ fuel LHV	Energy
	H2 LHV	33.3kWh/kg	source constant	Energy
	H2-equivalent requirement	104.1kg/day	3,467.5÷33.3	Supply
	Annual net demand	759,383kWh/yr	2,080.5×365	Energy
	PV capacity	1,500kW	1.5MW×1,000	Energy
	PV yield	1,100kWh/kW/yr	scenario benchmark	Energy
	PV loss	0.15	15%	Energy
	PV generation	1,402,500kWh/yr	1,500×1,100×0.85	Energy
	Annual PV ratio	1.85	1,402,500÷759,382.5	Annual energy only
	24h supply gate	Not proven by annual ratio	hourly power + storage + loss + weather model	Infra
	Supply loss	0.05 target	5%	Energy
	Supply stability	0.95 target	supplied ÷ required energy	Energy
Node	Node Delay	≤50ms	measured end-to-end node delay	Infra
	Dispatch Interval	5sec	60sec ÷ 12	Utilization
	Trips	73/day per node	4 nodes × 73 = 292/day; ceiling 480/day per node	Demand
	Fleet	25 vehicles	292÷25 = 11.7 trips/vehicle-day	Operation
	Load capacity	4pax	2×2 seats	Demand
	Energy per trip	28.5kWh/trip	2,080.5÷73	Energy
	Continuous ratio	1.00 target	24÷24	Utilization
	Availability	0.99 target	available time ÷ scheduled time	KPI
	Reliability	0.99 target	successful cycles ÷ total cycles	Stability
Vehicle	Control Sync	≤20ms	100ms ÷ 5	Motion sickness free
	Decision Sync	≤100ms	100ms	Motion sickness free
	Position Error	≤10cm	1m ÷ 10	Motion sickness free
	Platoon Stability	≤5%	0.05	Motion sickness free
	Control Cycle	100ms	1sec ÷ 10	Stability
	Response Delay	≤20ms	100ms ÷ 5	Motion sickness free
	Accel Jerk	≤0.5m/s²/s	1 ÷ 2	Motion sickness free
	Vibration	≤0.3Hz	1Hz ÷ 3.33	Motion sickness free
	Sync Accuracy	0.99	1 − 0.01	Stability
Control	Prediction Rate	1.00 target	successful predictions ÷ attempts	Operation
	Planning Rate	1.00 target	successful plans ÷ attempts	Operation
	Execution Rate	100Hz	1 ÷ 0.01sec	Control
	HPC Latency	≤10ms	measured inference latency	Stability
	Node Sync	≤50ms	measured synchronization delay	Stability
	Flow Capacity	40veh/h	60÷1.5	Infra
UX	Utilization Gain	0.10 target	0.75−0.65	KPI
Comparator	PEMFC efficiency	~60% direct H2	DOE technology comparison; not 22%	No exclusion on efficiency alone
	Battery downtime	4h/day assumption	24−20	Scenario only
	Battery utilization	0.83 assumption	20÷24	Scenario only
	ICE efficiency	0.30 assumption	scenario benchmark	Scenario only
Physics	Annual energy ratio	1.85	PV annual generation ÷ annual net demand	Energy
	Continuous supply	Validation pending	hourly balance required	Utilization
	Conversion loss	0.40	1−0.60	Energy
System	Feasibility gate	Not proven by multiplication	all independent safety, energy, operation, and financial gates must pass	Final
Conclusion	Hydrogen equivalent	104.1kg/day	2,080.5÷0.60÷33.3	Core
	Node operation	Validation pending	24h hourly model required	Core
	Utilization	0.75 target	18÷24	Core
	KPI status	Validation pending	not 0.58×0.75×0.70	Final

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Feasibility Conditions for Autonomous Mobility System

This page does not present financial outcomes. It defines scenario conditions that must be independently validated. Energy Efficiency = 0.58, Utilization = 0.75, and Load Factor = 0.70 are design targets, not derived facts. Feasibility depends on energy supply [kWh/day], available power [kW], storage duration [h], vehicle response [ms], control accuracy [cm], ODD evidence, and commercial constraints.

Powertrain alternatives must be compared on duty-cycle efficiency, degradation, refuelling/charging downtime, mass, packaging, infrastructure, and total cost. Direct-hydrogen PEM fuel cells are not excluded by a 22% efficiency assumption; DOE lists approximately 60% electrical efficiency for direct-H2 PEM. MS-SOFC remains the selected base-case candidate for this concept, subject to full TCO, durability, start-up, emissions, and ODD validation.

Node operation requires 2,080.5kWh/day of net traction energy. At 60% efficiency the fuel-LHV requirement is 3,467.5kWh/day, equivalent to 104.1kg-H2/day. PV generation of 1,402,500kWh/year gives an annual energy ratio of approximately 1.85, but continuous operation still requires an hourly supply-demand, storage, conversion-loss, and weather model.

"Motion sickness free" refers to a condition where control latency ≤20ms, decision latency ≤100ms, positioning error ≤10cm, and vibration ≤0.3Hz are simultaneously satisfied. This condition determines user acceptance and directly enables Utilization = 0.75 (=18h/day ÷ 24h/day). HPC maintains control at an execution rate of 100Hz (=1 ÷ 0.01sec), ensuring continuous satisfaction of response constraints. The definition is retained as the design requirement; compliance must be demonstrated by test evidence.

Block	Variable	Value / status	Formula / validation	KPI link
KPI	Energy Efficiency	0.58 target	metered useful output ÷ purchased input	Energy
	Utilization	0.75 target	18÷24	Operation
	Load Factor	0.70 target	2.8÷4	Demand
Constraint	Operation Hours	24h/day target	operating window	Utilization
	Stability	0.95 target	1−0.05	Energy
	Throughput	40veh/h	1 vehicle ÷1.5min	Infra
Node	Net energy	2,080.5kWh/day	28.5×73	Energy
	Fuel input	3,467.5kWh/day	2,080.5÷0.60	Energy
	SOFC efficiency	0.60 target	net electricity ÷ fuel LHV	Energy
	H2 LHV	33.3kWh/kg	constant	Energy
	H2 required	104.1kg/day	3,467.5÷33.3	Supply
	Annual net energy	759,383kWh/yr	2,080.5×365	Energy
	PV capacity	1,500kW	1.5MW×1,000	Energy
	PV yield	1,100kWh/kW/yr	benchmark assumption	Energy
	PV loss	0.15	15%	Energy
	PV generation	1,402,500kWh/yr	1,500×1,100×0.85	Energy
	Annual PV ratio	1.85	1,402,500÷759,382.5	Annual only
	Continuous supply	Not yet demonstrated	hourly balance + storage + reserve	Utilization
	Supply stability	0.95 target	supplied ÷ required	Energy
	Node delay	≤50ms target	measured	Infra
	Dispatch interval	5sec	60÷12	Utilization
	Trips	73/day per node	4 nodes × 73 = 292/day	Demand
	Fleet	25	292÷25 = 11.7/vehicle-day	Operation
	Load	4pax	2×2	Demand
	Energy/trip	28.5kWh	2,080.5÷73	Energy
	Reliability	0.99 target	successful cycles ÷ total	Stability
	Availability	0.99 target	available ÷ scheduled time	KPI
Vehicle	Control Sync	≤20ms	100ms÷5	Motion sickness free
	Decision Sync	≤100ms	100ms	Motion sickness free
	Position Error	≤10cm	1m÷10	Motion sickness free
	Platoon Stability	≤5%	0.05	Motion sickness free
	Control Cycle	100ms	1sec÷10	Stability
	Response	≤20ms	100ms÷5	Motion sickness free
	Jerk	≤0.5m/s²/s	1÷2	Motion sickness free
	Vibration	≤0.3Hz	1Hz÷3.33	Motion sickness free
	Sync Accuracy	0.99	1−0.01	Stability
Control	Prediction	1.00 target	successful prediction ratio	Operation
	Planning	1.00 target	successful plan ratio	Operation
	Execution	100Hz	1÷0.01sec	Control
	HPC latency	≤10ms target	measured	Stability
	Node sync	≤50ms target	measured	Stability
	Flow	40veh/h	60÷1.5	Infra
UX	Utilization Gain	0.10 target	0.75−0.65	KPI

This defines the conditions under which autonomous mobility may become operationally feasible. Passing one ratio does not substitute for validation of the remaining gates.

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Proposed Responsibility Structure for Autonomous Mobility System

This is a proposed RACI hypothesis for discussion, not a statement of current accountability of any specific company or consortium. Final responsibility must be confirmed by corporate governance, contracts, permits, product-liability allocation, operational rules, and the approved safety case.
Continuous operation depends on explicit ownership across orchestration, fleet operation, vehicle/system safety, node execution, energy supply, commercial settlement, and finance. A method such as model-based development is not an accountable legal entity; each task must be assigned to an organization or contracted operator.

Domain	Function / task	Accountable (A)	Responsible (R)	Support / consulted	Status
Demand	Generate demand	Mobility Platform	Mobility Platform	AD Platform / partners	Proposed
	Capture demand	Mobility Platform	Mobility Platform	MaaS Operator	Proposed
	Set pricing	Commercial owner	Commercial owner	Finance / regulator	Contract required
	Approve pricing	Vehicle OEM / business owner	Commercial owner	Legal / regulator	Approval required
	Optimize load	MaaS Operator	Mobility Platform	AD Platform	Proposed
	Manage UX	Mobility Platform	Mobility Platform	AD Platform / users	Proposed
	Define service	Vehicle OEM / business owner	Mobility Platform	MaaS Operator / user	Proposed
Operation	Execute dispatch	MaaS Operator	MaaS Operator	Mobility / AD Platforms	Permit + SLA
	Operate fleet	MaaS Operator	MaaS Operator	OEM / AD Platform	Permit + SLA
	Maintain service	MaaS Operator	MaaS Operator	Node operator / Mobility Platform	SLA
	Manage utilization	MaaS Operator	Mobility Platform	AD Platform	Proposed
	Define hours	MaaS Operator	MaaS Operator	Regulator / node operator	Permit
	Control dispatch	MaaS Operator	AD control platform	Mobility Platform	Authority boundary
Vehicle	Define architecture	Vehicle OEM	Vehicle OEM	AD Platform / Tier-1	Product governance
	Supply platform	Vehicle OEM	Vehicle OEM	Tier-1 suppliers	Commercial contract
	Supply drivetrain	Vehicle OEM	Tier-1 supplier(s)	AD Platform	Commercial contract
	Package cabin	Vehicle OEM	Body integrator / supplier	Mobility Platform	Commercial contract
	Maintain vehicle	Fleet owner / MaaS Operator	Certified service provider	Vehicle OEM	SLA
	Realize control	Vehicle OEM	AD Platform + Tier-1	MaaS Operator	Safety case
	Maintain position	AD Platform / OEM	AD Platform	Mobility Platform	Safety case
Facility	Provide site	Node / Parking Operator	Node / Parking Operator	Vehicle OEM	Contract
	Operate facility	Node / Parking Operator	Node / Parking Operator	MaaS Operator	Permit + SLA
	Sell services	Commercial owner	Mobility Platform / provider	Node operator	Contract
	Use platform	MaaS Operator	MaaS Operator	AD Platform	SLA
	Manage circulation	Node operator	Node operator	Vehicle OEM	SLA
Service	Provide booking	Mobility Platform	Mobility Platform	AD Platform	Service governance
	Integrate service	Mobility Platform	Mobility Platform	Service providers	API contract
	Sell package	Commercial owner	Commercial owner	Finance	Contract
	Optimize stays	MaaS Operator	Mobility Platform	Service providers	Proposed
	Operate service	Service provider	Service provider	MaaS Operator	SLA
MaaS	Operate platform	MaaS Operator	AD + Mobility Platforms	Vehicle OEM	Authority boundary
	Use platform	MaaS Operator	MaaS Operator	AD Platform	SLA
External	Provide entry	External platform owner	External platform owner	Mobility Platform	API contract
	Send traffic	External platform owner	External platform owner	MaaS Operator	Commercial contract
	Receive traffic	MaaS Operator	Mobility Platform	External platform	Commercial contract
Energy	Define supply	Energy business owner	Energy integrator / supplier	AD Platform / node operator	Contract
	Produce hydrogen	Energy supplier	Energy supplier	Node operator	Permit + quality
	Distribute energy	Energy supplier	Energy integrator / supplier	Node operator	Contract
	Manage logistics	Energy Integrator	Energy Integrator	Energy supplier	Contract
	Operate energy	Node operator	Node operator	AD Platform	SLA
Software	Define scope	Vehicle OEM / product owner	AD Platform	Mobility Platform	Product governance
	Develop control	AD Platform	AD Platform	Vehicle OEM / Tier-1	Release governance
	Operate compute	AD Platform	Cloud / edge operator	MaaS Operator	SLA
	Maintain dispatch	MaaS Operator	AD + Mobility Platforms	Node operator	SLA
Finance	Collect mobility	Commercial owner	Finance / payment operator	Mobility Platform	Settlement contract
	Collect facility	Node operator	Finance / payment operator	Mobility Platform	Settlement contract
	Collect energy	Energy business owner	Finance / payment operator	Energy Integrator	Settlement contract
	Maintain projection	Business owner	Finance	Vehicle OEM	Governance
	Collect data	Data controller	Mobility Platform	AD Platform / operators	Privacy + data contract
	Collect facility data	Node operator	Mobility Platform	AD Platform	Privacy + data contract

Responsibility is complete only after the accountable legal entity, operational authority, safety authority, data controller, and revenue/cost owner are all confirmed.

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Pricing Structure and Billing Rules for Autonomous Mobility System

The amounts below remain illustrative scenario tariffs, not verified market prices. To prevent double counting, one base trip fare is selected per trip; add-ons apply only when the corresponding service is consumed; subscriptions offset eligible base charges; B2B platform fees are charged only when not already embedded in another contract; dividends/revenue shares are settlements below revenue, not customer prices.

Category	Payer	Payee	Charge	Billing rule	Illustrative formula / annualization	P&L mapping
User	User	Service provider	Facility Fee	Add-on	¥500/visit×200/day×365 = ¥36.500M/yr	MaaS service
			Premium Service	Add-on	¥1,500/booking×120/day×365 = ¥65.700M/yr	Service revenue
			Add-on	Add-on	¥600/use×200/day×365 = ¥43.800M/yr	Service revenue
		MaaS Operator	Service Fee	Base fare (exclusive)	¥1,000/trip×130/day×365 = ¥47.450M/yr	Trip revenue
			Trip Fee	Base fare (exclusive)	¥2,000/trip×100/day×365 = ¥73.000M/yr	Trip revenue
			Premium Trip	Base fare (exclusive)	¥2,500/trip×90/day×365 = ¥82.125M/yr	Trip revenue
			Subscription	Offsets eligible base fares	¥15,000/user×1,200 users = ¥18.000M/yr	Subscription revenue
			Night Trip	Base fare (exclusive)	¥2,700/trip×80/day×365 = ¥78.840M/yr	Trip revenue
		Parking Operator	Premium Facility	Add-on	¥800/visit×120/day×365 = ¥35.040M/yr	Facility revenue
			Loyalty Fee	Add-on / retention	¥200/visit×185/day×365 = ¥13.505M/yr	Service revenue
Facility	AD Platform	Parking Operator	Site Rent	B2B fixed fee	¥300,000/month×30×12 = ¥108.000M/yr	Node OPEX
			Operating Access	B2B access fee	¥700/visit×90/day×365 = ¥22.995M/yr	Node OPEX
	Parking Operator	Service Fee	B2B service	¥300/visit×200/day×365 = ¥21.900M/yr	Node revenue	
	Service Provider	Parking Operator	Upgrade Rent	B2B fixed fee	¥2,000,000/month×30×12 = ¥720.000M/yr	Node OPEX
Service	Service Provider	Mobility Platform	Booking Fee	Per booking	¥100/booking×150/day×365 = ¥5.475M/yr	Reservation fee
			CRM Fee	B2B fixed fee	¥80,000/month×30×12 = ¥28.800M/yr	Platform fee
			API Fee	B2B usage fee	¥80/call×250/day×365 = ¥7.300M/yr	Platform fee
			Premium UX	B2B product fee	¥250/booking×180/day×365 = ¥16.425M/yr	Platform fee
		AD Platform	Optimization Fee	B2B product fee	¥50/call×300 calls/day×365 = ¥5.475M/yr	Platform fee
			Premium API	B2B usage fee	¥150/call×60/day×365 = ¥3.285M/yr	Platform fee
			Sync Fee	B2B product fee	¥12/call×5,500/day×365 = ¥24.090M/yr	Platform fee
		Service Provider	Late Fee	Customer charge	¥1,200/use×70/day×365 = ¥30.660M/yr	Service revenue
Mobility	MaaS Operator	AD Platform	Control Fee	B2B product fee	¥100/dispatch×100/day×365 = ¥3.650M/yr	Control OS
			Routing Fee	B2B product fee	¥120/dispatch×110/day×365 = ¥4.818M/yr	Platform fee
		Mobility Platform	UX Fee	B2B product fee	¥50/trip×100/day×365 = ¥1.825M/yr	Platform fee
		Parking Operator	Advanced Control	B2B service fee	¥150/dispatch×90/day×365 = ¥4.928M/yr	Node service
	AD Platform	MaaS Operator	Incentive	Contra-revenue / settlement	¥200,000/month×30×12 = ¥72.000M/yr	Settlement; not gross revenue
	User	MaaS Operator	Surge Pricing	Base fare adjustment	¥2,800/trip×80/day×365 = ¥81.760M/yr	Trip revenue
			Off-peak Discount	Base fare adjustment	¥1,600/trip×70/day×365 = ¥40.880M/yr	Trip revenue
Platform	MaaS Operator	AD Platform	OS Fee	B2B fixed fee	¥80,000/month×30×12 = ¥28.800M/yr	Platform fee
			Monthly OS	B2B fixed fee	¥250,000/month×30×12 = ¥90.000M/yr	Platform fee
	Service Provider	AD Platform	Node Fee	B2B fixed fee	¥200,000/month×30×12 = ¥72.000M/yr	Node service
			Data Fee	B2B usage fee	¥100,000/month×30×12 = ¥36.000M/yr	Platform fee
	Parking Operator	AD Platform	Facility Fee	B2B fixed fee	¥200,000/month×30×12 = ¥72.000M/yr	Node service
			Analytics Fee	B2B fixed fee	¥120,000/month×30×12 = ¥43.200M/yr	Platform fee
			Control OS	B2B product fee	¥220/trip×110/day×365 = ¥8.833M/yr	Control OS
			Booking Link	B2B usage fee	¥180,000/month×30×12 = ¥64.800M/yr	Reservation fee
External	External Platform	AD Platform	API Fee	B2B usage fee	¥120/use×5,000/day×365 = ¥219.000M/yr	Platform fee
			IoT Fee	B2B usage fee	¥90/booking×200/day×365 = ¥6.570M/yr	Platform fee
			Tx Fee	B2B usage fee	¥200/booking×100/day×365 = ¥7.300M/yr	Platform fee
			Premium API	B2B usage fee	¥150/call×6,000/day×365 = ¥328.500M/yr	Platform fee
			Premium Tx	B2B usage fee	¥250/booking×180/day×365 = ¥16.425M/yr	Platform fee
			Premium Referral	Revenue share / settlement	15% of eligible booking revenue	Settlement; not gross revenue
			Sync Fee	B2B usage fee	¥12/call×5,500/day×365 = ¥24.090M/yr	Platform fee
Energy	AD Platform	Finance Co.	Energy Fee	B2B service	20kWh/trip×100/day×¥30×365 = ¥21.900M/yr	Energy revenue
			Peak Energy	B2B service	25kWh/trip×90/day×¥45×365 = ¥36.956M/yr	Energy revenue
			Smart Energy	B2B service	¥70/use×120/day×365 = ¥3.066M/yr	Energy revenue
	Service Provider	AD Platform	Energy Opt	B2B product fee	¥50/use×150/day×365 = ¥2.738M/yr	Platform fee
Finance	MaaS Operator	Finance Co.	Base Energy	B2B settlement	18kWh/trip×110/day×¥28×365 = ¥20.236M/yr	Energy OPEX
			Lease	B2B lease	¥3,000,000/year×365-day basis = contract specific	Lease / CAPEX
			Dividend	Settlement only	profit distribution after tax	Not revenue
			Node Lease	B2B lease	¥4,500,000/year per node	Lease / CAPEX
			Node Lease	B2B lease	¥2,500,000/year per node	Lease / CAPEX
	Service Provider	Finance Co.	Lease	B2B lease	¥2,000,000/year×365-day basis = contract specific	Lease / CAPEX
			Dividend	Settlement only	profit distribution after tax	Not revenue
			Service Lease	B2B lease	¥2,500,000/year per asset	Lease / CAPEX

Only lines explicitly mapped to the P&L may be included. Settlement lines and mutually exclusive base fares must not be added again.

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Autonomous Mobility Business Control Structure

System conditions define the feasibility of control and set operational boundaries. Values below are phase targets or scenario assumptions; they are not externally verified outcomes. Minimum gates on P1 and phase targets here are different by design and must be labelled separately.
Revenue uses unit-consistent formulas: annual passenger revenue = Nodes × Trips/day per node × Seats × Load factor × Blended fare × 365. Break-even trips/day cannot be calculated without a defined annual controllable-cost allocation.

Layer	Control item	Decision logic	2030	2035	P&L impact
Demand	Demand generation	Requests ÷ potential demand	≥0.50	≥0.70	Trips
	Conversion	Bookings ÷ requests	≥0.23	≥0.35	Trips
	Matching	Matched ÷ requests	≥0.90	≥0.95	Trips
	Trip realization	Executed ÷ matched	≥0.95	≥0.95	Trips
	Stabilization	Matched trips + planned demand	≥0.80	≥0.95	Trips
	Time synchronization	Calendar-based demand	≥0.60	≥0.85	Trips
	Service capture	Reservation-based trips ÷ total	≥0.50	≥0.75	Trips
	Demand-cap control	Demand ≤ system capacity	100%	100%	Trips cap
Operation	Trips	Executed trips/day per node	73	140	Trips
	Utilization	Operating time ÷ 24h	≥0.75	≥0.85	Trips
	Dispatch success	Successful dispatch ÷ requests	≥0.95	≥0.99	Trips
	Supply-body cycle	Active body ÷ total cycle	≥0.80	≥0.95	Utilization
	Waiting time	Passenger request-to-pickup	≤360sec	≤180sec	Trips
	Turnaround	Idle + service time	≤25min	≤10min	Utilization
	Queue control	Total wait ÷ requests	≤300sec	≤120sec	Conversion
	Optimization	Served ÷ queued	≥0.90	≥0.98	Trips
	Scheduled continuity	Continuous-operation ratio	≥0.80	≥0.95	Utilization
	Operation cap	Operation ≤24h	Fixed	Fixed	Cap
Vehicle	Latency control	Control delay	≤20ms	≤10ms	Dispatch
	Stability	Deviation ÷ target	≤5%	≤5%	Safety
	Intervention	Interventions ÷ km	≤0.010/km	≤0.0001/km	Utilization
	Motion response	Control latency / planning / ride targets	Design gate	Design gate	Utilization
	Response accuracy	Correct ÷ total control	≥0.90	≥0.98	Dispatch
	Platoon composition	Vehicles per formation	≥1.5	≥3.0	Load
	Ride continuity	Continuous ride + trips	≥0.85	≥0.90	Utilization
	Thermal constraint	Temperature limit	Fixed	Improved	Operation
	Vehicle availability	Available fleet ÷ total	≥0.80	≥0.95	Trips
Energy	End-to-end efficiency	Useful output ÷ purchased input	0.58 target	0.65 target	Cost
	Energy cost/trip	Cost ÷ trips	≤¥500	≤¥350	Cost
	Supply stability	Continuous supplied ÷ required	≥0.95	≥0.99	Utilization
	H2-equivalent requirement	Net energy ÷ 0.60/0.70 ÷33.3	104.1kg/day	171.2kg/day	Trips cap
	PV role	Annual energy contribution	Annual ratio only	Annual ratio only	Cost
	Storage capacity	Hourly storage / reserve	Required	Required	Trips cap
	Distribution	Loss + supply	≤10%	≤5%	Cost
	Node allocation	Allocated ÷ demand	≥0.90	≥0.98	Trips
	Energy scheduling	Scheduled + required	≥0.85	≥0.95	Trips
System	Trip generation	Demand×Matching×Dispatch×Utilization	73/day	140/day	Revenue
	Load optimization	Passengers ÷ capacity	0.70	0.80	Revenue
	Fare	Day / night fare (night 35% / 60%)	¥2,200 / ¥2,700	¥2,000 / ¥2,700	Revenue
	Throughput	Trips ÷ hour	≥1.5	≥3.0	Revenue
	Density	Trips ÷ area	Low	High	Revenue
	Scaling factor	Trips 2035 ÷ 2030	1.00×	1.92×	Revenue
	Energy linkage	28.5kWh × trips per node	2,080.5kWh	3,990kWh	Cost
	Hydrogen linkage	Energy ÷ 0.60/0.70 ÷33.3	104.1kg	171.2kg	Cost
	Trips/vehicle-day	Nodes × Trips ÷ Fleet	11.7	10.1	Trips cap
	Congestion	Load ÷ capacity	Medium	High	Limit
	Node capacity	Throughput ≤ capacity	Limited	Expanded	Cap
Financial	Annual passenger revenue	Nodes×Trips×Seats×Load×Fare×365	¥0.709B	¥17.807B	Profit
	Revenue per vehicle trip	Seats×Load×Blended fare	¥6,650	¥9,680	Revenue
	Contribution	Allocated revenue − allocated cost	See rates	See rates	IRR
	Break-even trips/day	Annual controllable cost ÷ (Seats×Load×Fare×365)	TBD	TBD	Profit
	Cost control	Energy + operation + vehicle + infrastructure	High	Reduced target	Cost
	Price constraint	Applicable fare / contract rules	Validate	Validate	Revenue
	IRR hurdle	Return target	≥10%	≥12%	Investment

This defines unit-consistent control logic connecting operational conditions to trips, revenue, cost, and investment gates.

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Autonomous Mobility Business Profit & Loss Structure (2030 / 2035)

This is a scenario P&L, not a forecast. Each monetary line is recalculated from the displayed formula. Non-monetary operating drivers are separated from revenue. The prior ¥36B node-sale line is excluded from the base case because asset-transfer price, delivery cost, and COGS were not defined; recurring node-operation service remains included.

Layer	Category	Item	2030 formula	2030	2035 formula	2035
Revenue	Energy	Elec Internal	4.0M×20%×50%×1,200×¥15	¥7.200B	4.0M×30%×50%×1,500×¥15	¥13.500B
		Elec External	4.0M×25%×50%×1,200×¥15	¥9.000B	4.0M×35%×50%×1,500×¥15	¥15.750B
		H2 Internal	4.0M×80%×1%×67×¥333	¥0.714B	4.0M×70%×20%×67×¥333	¥12.494B
		H2 External	4.0M×20%×1%×67×¥333	¥0.178B	4.0M×30%×20%×67×¥333	¥5.355B
		Titanium	80,000×0.3×¥72,000	¥1.728B	same	¥1.728B
		HPC + quantum	600 GPU + 1 quantum	¥2.076B	5,400 GPU + 12 quantum	¥31.378B
Driver		24H Utilization	18÷24	0.75	20.4÷24	0.85
		Load Factor	2.8÷4	0.70	4.0÷5	0.80
		Swap-body Util.	Active÷Total	0.80	Dynamic composition	0.95
		Night Operation	Night trips÷Trips	0.35	Continuous operation	0.60
		Auto Loading Rev.	Loading fee rev.÷Total	0.20	Integrated logistics	0.09
Revenue	MaaS	Passenger Trips	4×73×2.8×¥2,375×365	¥0.709B	36×140×4.0×¥2,420×365	¥17.807B
		Night Cargo	25×24×¥5,000×365×16.6%	¥0.182B	500×24×¥5,000×365×34.6%	¥7.577B
		Mobile Storage	400×¥40,000×12	¥0.192B	5,000×¥40,000×12	¥2.400B
	Platform	Auto Loading	269,000×¥22,300	¥5.999B	538,000×¥22,300	¥11.997B
		Platform Fee	Dispatch/API fee	¥0.800B	Integrated OS fee	¥6.500B
		Reservation Fee	Booking linkage	¥0.350B	Integrated booking	¥2.200B
		Node Service	Node operation fee	¥0.420B	Node ecosystem service	¥4.800B
Total		Total Revenue	sum monetary revenue	¥29.548B	sum monetary revenue	¥133.486B
OPEX	Energy	H2 Internal COGS	4.0M×80%×1%×67×¥150	▲¥0.322B	4.0M×70%×20%×67×¥150	▲¥5.628B
		H2 External COGS	4.0M×20%×1%×67×¥150	▲¥0.080B	4.0M×30%×20%×67×¥150	▲¥2.412B
		Elec Internal	4.0M×20%×50%×1,200×¥6.7	▲¥3.216B	4.0M×30%×50%×1,500×¥6.7	▲¥6.030B
		Elec External	4.0M×25%×50%×1,200×¥6.7	▲¥4.020B	4.0M×35%×50%×1,500×¥6.7	▲¥7.035B
		Titanium	80,000×0.3×¥6,000	▲¥0.144B	same	▲¥0.144B
		Material Cycle	scenario cost	▲¥0.300B	onboard circulation	▲¥1.800B
		Node Optimization	distributed control	▲¥0.120B	integrated mesh	▲¥0.950B
		Energy Sync	In/Out sync	▲¥0.080B	real-time optimization	▲¥0.600B
	MaaS	MaaS Ops	25×¥81.0M	▲¥2.025B	500×¥45.6M	▲¥22.800B
		Dispatch Ops	fleet dispatch	▲¥0.550B	integrated dispatch	▲¥4.200B
		Remote Ops	remote support	▲¥0.300B	integrated operation	▲¥2.100B
	Platform	Platform Ops	¥1.0B/year	▲¥1.000B	¥6.0B/year	▲¥6.000B
		Cloud Ops	base + compute ops	▲¥0.952B	base + compute ops	▲¥10.284B
		AI Ops	AI orchestration	▲¥0.220B	integrated AI control	▲¥2.400B
Total		Total OPEX	sum	▲¥13.329B	sum	▲¥72.383B
SG&A	Energy	Energy SG&A	energy product revenue×10%	▲¥1.882B	energy product revenue×10%	▲¥4.883B
		Maintenance	scenario	▲¥0.250B	integrated maintenance	▲¥1.800B
	MaaS	MaaS SG&A	MaaS revenue×15%	▲¥0.162B	MaaS revenue×15%	▲¥4.168B
		Customer Ops	user support	▲¥0.180B	integrated support	▲¥1.500B
	Platform	Platform SG&A	SW revenue×20%	▲¥1.514B	SW revenue×20%	▲¥5.099B
		Partner Ops	alliance operation	▲¥0.120B	integrated alliance	▲¥1.100B
Total		Total SG&A	sum	▲¥4.108B	sum	▲¥18.550B
Profit	Total	EBITDA	Revenue−OPEX−SG&A	¥12.111B	Revenue−OPEX−SG&A	¥42.553B
Dep.	Energy/MaaS/Platform	Node	9.600B÷15y	▲¥0.640B	86.400B÷15y	▲¥5.760B
		GTFC	18.450B÷15y	▲¥1.230B	22.600B÷15y	▲¥1.507B
		Mill	0.600B÷10y	▲¥0.060B	4.800B÷10y	▲¥0.480B
		Battery	5.000B÷10y	▲¥0.500B	6.400B÷10y	▲¥0.640B
		Magnetic	3.000B÷10y	▲¥0.300B	3.800B÷10y	▲¥0.380B
		HTS	4.000B÷10y	▲¥0.400B	5.100B÷10y	▲¥0.510B
		Water	2.000B÷10y	▲¥0.200B	2.600B÷10y	▲¥0.260B
		HPC / quantum	5.0B÷15y+1.0B÷7y	▲¥0.476B	45.0B÷15y+12.0B÷7y	▲¥4.714B
		Energy Device	Included in Mill/Batt/HTS	¥0.000B	Included in Mill/Batt/HTS	¥0.000B
		Vehicle Dev	(50.000−8.333)B÷5y	▲¥8.333B	(63.900−50.000)B÷5y	▲¥2.780B
		Prototype	(10.000−2.500)B÷5y	▲¥1.500B	fully depreciated by 2032	¥0.000B
		Swap-body Dev	Included in Vehicle Dev	¥0.000B	Included in Vehicle Dev	¥0.000B
		Platform Dev	(40.000−6.667)B÷5y	▲¥6.667B	(59.400−40.000)B÷5y	▲¥3.880B
		AI Platform	Included in Platform Dev	¥0.000B	Included in Platform Dev	¥0.000B
		Control OS	Included in Platform Dev	¥0.000B	Included in Platform Dev	¥0.000B
		Infra Sync	Included in Platform Dev	¥0.000B	Included in Platform Dev	¥0.000B
		Revenue Logic	Included in Platform Dev	¥0.000B	Included in Platform Dev	¥0.000B
Total		Total Depreciation	sum	▲¥20.306B	sum	▲¥20.911B
Profit	Total	Operating Profit	EBITDA−Dep.	▲¥8.195B	EBITDA−Dep.	¥21.642B
		Contribution before depreciation	Revenue−OPEX−SG&A	¥12.111B	Revenue−OPEX−SG&A	¥42.553B
		Operating Margin	Operating Profit÷Revenue	-27.7%	Operating Profit÷Revenue	16.2%
		Project Payback	see cash-flow table	not achieved	see cash-flow table	discounted not achieved by 2035
		Project IRR	reconciled cash flow	-36.49%	12% hurdle	not met

Assumptions retained unless corrected above: city 1/5; MaaS vehicles 25/500; nodes 4/36; passenger cars 4.0M/year; trucks 80,000/year; electricity sales ¥15/kWh and cost ¥6.7/kWh; H2 sales ¥333/kg and cost ¥150/kg; titanium exchange 80,000×0.3, sales ¥72,000 and cost ¥6,000. These are scenario assumptions requiring primary-source validation.

Depreciation follows in-service vintages: ¥148.65B is in service at 2030 and ¥322.000B at 2035 (node ¥2.4B, quantum ¥1.0B each); ¥75.35B of 2025-2030 CAPEX is construction in progress placed in service 2031-2035; 5-year assets in service by 2030 are fully depreciated by 2034.

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2030 Vehicle Platform

A 12m autonomous MaaS skateboard uses an MS-SOFC scenario with city gas 70% and mill-derived hydrogen 30%. Hydrogen is supplied from stationary/node-side milling; batteries support transient power. Component ranges sum to approximately 9.8–15.1t before the stated 10–15% integration reserve; the upper case approaches 17.3t. Therefore, payload is not claimed until body, fluids, passengers, wiring/piping, axle loads, and certified curb mass are included.

The 10–15kg LH2 buffer requires approximately 340L after the 1.6 packaging factor. The envelope is corrected to 450×550×1,400mm (346.5L external envelope). For 240kW electrical output at 60% efficiency, fuel input is 400kW and nominal rejected heat is approximately 160kW, not 96kW.

Component	Qty	Performance	Formula	Corrected result	H/W/D mm	Weight	Layout
1 Energy/H2/Power		12m flat skateboard	-	upper container unused	-	-	overall
Total energy demand	1	1.2kWh/km	1.2×1,000	1,200kWh	-	-	overall
Required fuel energy	1	MS-SOFC 60%	1,200÷0.60	2,000kWh-LHV	-	-	overall
Hydrogen ratio	1	H2 30% blend	2,000×0.30	600kWh-H2	-	-	overall
Required H2	1	33.3kWh/kg	600÷33.3	18.0kg-H2	-	-	overall
City-gas ratio	1	70%	2,000×0.70	1,400kWh-gas	-	-	overall
LH2 buffer	1	10–15kg	reserve	10–15kg-H2	450/550/1400	150–250kg	left
LH2 installed volume	1	insulation×1.6	15÷70.8×1.6	≈340L	450/550/1400	included	left
City-gas tank	2	blended-fuel support	-	operation reserve	300/400/1600	300–500kg	left
Water tank	2	cooling/process water	-	≈180L	250/400/1200	180kg	left
Process-water control	2	steam-to-carbon / condensate	validated setpoint	not NOx claim	180/250/400	80–120kg	front
O2 membrane	2	optional O2 management	test gate	efficiency TBD	180/250/450	80–120kg	front
CO2 membrane	2	reformate cleanup	70–90% target	CO2 removal target	220/250/500	100–150kg	rear
Steam membrane	2	H2O separation	test gate	reforming support	180/250/450	80–120kg	rear
Reformer	2	steam reforming	fuel-path model	H2+CO reformate	260/350/700	250–350kg	front
Gas/liquid separator	2	50L/min	50×2	100L/min	220/250/450	120–180kg	F/R
H2 purifier	2	50L/min	50×2	100L/min	220/250/500	120–180kg	F/R
Vaporizer	2	20kg/h	20×2	40kg/h	240/300/600	120–180kg	left
LH2 pump	2	50L/min	50×2	100L/min	180/220/350	60–100kg	left
Pressure regulator	4	high→low P	-	4 systems	120/200/300	40–60kg	overall
Shutoff valve	6	safety shutoff	-	6 systems	120/180/250	40–60kg	overall
Vent line	2	safety vent	-	2 systems	100/100/1400	20–40kg	upper
Safety sensors	1	H2/P/T monitor	-	integrated	80/200/300	<30kg	overall
2 Electric/Thermal		MS-SOFC power control	-	battery transient	-	-	center
MS-SOFC modules	6	40kW/unit	40×6	240kW	320/1000/1400	1,200–1,800kg	center
MS-SOFC efficiency	1	electrical efficiency	-	≈60% target	-	-	center
Inverter	2	120kW	120×2	240kW	180/420/650	180–260kg	upper center
DC/DC	2	100kW	100×2	200kW	160/350/500	120–180kg	upper center
Solid-state battery	1	400kWh	-	400kWh	260/1200/2200	2,500–3,500kg	center
HV J-BOX	1	HV distribution	-	1 system	180/350/500	40–80kg	center
EMS/VCU	1	energy+vehicle ctrl	-	1 system	120/250/300	40–80kg	center
Radiator	2	rejected heat	240×(1/0.60−1)	≈160kW	180/500/1400	180–300kg	front
Cooling pump	2	50L/min	50×2	100L/min	160/180/250	40–80kg	overall
Waste-heat recovery	1	Rankine/thermal	validated recovery	aux. power TBD	260/600/900	250–450kg	rear
Insulation monitoring	1	HV safety	-	1 system	80/200/300	<20kg	center
Power protection	1	fuse/breaker	-	integrated	80/200/300	<20kg	center
3 AD Control		autonomous execution	-	AD operation	-	-	upper
AD computer	1	prediction/planning	latency≤100ms	control target	120/400/500	40–80kg	upper
Sensor suite	1	LiDAR/camera/radar	-	360° sensing	120/500/800	40–80kg	upper
Fusion/localization	1	map+IMU+GNSS	-	position control	80/300/400	<30kg	upper
ODD/DDT monitor	1	state-dependent safety gate	condition class	HOLD / restrict / MRM / STOP	80/250/300	<20kg	center
Dispatch/routing ECU	1	trip execution	-	fleet control	100/300/400	20–40kg	upper
ETA/latency control	1	synchronization	latency≤50ms	AD stability target	80/250/300	<20kg	upper
4 Drivetrain		electric drive	-	AD ride stability	-	-	corners
IWM	4	80kW/wheel	80×4	320kW	320/500/500	1,000–1,400kg	corners
Regen brake	4	regenerative	-	4 wheels	400/400/400	120–180kg	corners
Air brake	4	braking	-	4 wheels	400/400/400	120–180kg	corners
Knuckle+EPS	4	steer support	-	4 wheels	260/320/420	250–350kg	corners
Air suspension+CFRP	4	vibration suppression	-	4 wheels	260/350/450	250–350kg	corners
CTIS tire	4	pressure control	-	4 wheels	220/500/500	included	corners
Ride comfort target	1	no motion sickness	lateral accel control	UX stabilization	-	-	cabin
5 Structure		flat payload frame	-	550–600mm target	-	-	overall
Crossmember	6	structural beam	-	6 beams	160/1600/160	250–400kg	lateral
MH main chassis	2	H2 storage+frame	-	≈10kg-H2	400/350/5500	400–800kg	longitudinal
LH2 subframe	2	cooling+support	-	2 systems	260/450/2200	300–500kg	lower
Protection/firewall	1	impact+fire safety	-	integrated	50–100/car	400–700kg	lower/center
Container lock	4	swap fixation	-	4 points	180/300/300	80–120kg	upper
Swap positioning	4	±mm alignment	-	4 points	180/200/200	80–120kg	upper
Service interface	1	refill/drain/inspect	-	gas/LH2/water	150/500/900	<120kg	L/R
Boarding assist	1	kneel+2-step	−100mm	access assist	220/300/800	120–200kg	right
Payload margin	1	25t GVW−verified curb mass	full integration required	TBD; prior 10–13t claim removed	-	-	overall
Service margin	1	maintenance reserve	total×10–15%	reserve	-	included	overall

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2035 Vehicle Platform

A 12m platform targets a 380–400mm low floor. The platform-weight and 13–16t payload claims are removed pending full integrated mass, body, fluids, passengers, wiring/piping, axle-load, and certification data. The energy balance now separates the 400kWh air-battery contribution from PCFC energy demand.

PCFC-only hydrogen demand is 48.0kg/day (=1,200÷33.3÷0.75). Net of the 400kWh air battery, plus 178kWh of regen lost to the mills and 287kWh to release H2 from the MH at 0.21V, the PCFC must deliver 1,265kWh: 50.6kg/day of H2. The six mills (rated 64.9kg/day) get only 0.03kg/day from braking energy, so Node refuelling supplies 50.6kg/day (3 stops of 20kg). The 10-15kg LH2 buffer needs ≈340L (350×850×1,200mm, 357L). Peak heat is ≈108kW: PCFC 80, mill 19, drive 9kW; the MH absorbs 12kW.

Component	Qty	Performance	Formula	Corrected result	H/W/D mm	Weight	Layout
1 Energy/H2/Power		system demand	1.2×1,000	1,200kWh	-	-	overall
Gross H2 before air battery	1	PCFC 75%	1,200÷(33.3×0.75)	48.0kg-H2	-	-	overall
Air-battery energy	4	100kWh/unit	100×4	400kWh	120/350/1200	≈849kg	right
H2 reduction equivalent	1	25kWh/kg-H2	400÷25	16.0kg-H2 eq.	-	-	right
Net PCFC energy	1	net + regen + MH	800+178+287	1,265kWh	-	-	overall
Net H2 required	1	PCFC 75%	1,265÷(33.3×0.75)	50.6kg-H2	-	-	overall
Hydrogen generation	6	0.865kg/h	238kWh÷8,305	0.03kg (64.9 rated)	240/450/650	720–1,080kg	F/R
Node refuel	1	3 stops × 20kg MH	50.6−0.03	50.6kg-H2	-	-	overall
LH2 buffer	1	10–15kg reserve	-	10–15kg-H2	350/850/1200	100–150kg	left
LH2 installed volume	1	insulation×1.6	15÷70.8×1.6	≈340L	350/850/1200	included	left
Water tank	6	9kg/kg-H2 gross	64.9×9	584L	200/400/1200	584kg	left
Slurry tank	6	water×1.4	584×1.4	818L	230/450/1300	980–1,227kg	left
Titanium cartridge	6	3kg/0.865kg-H2	64.9÷0.865×3	≈225kg	160/300/800	225kg	L/R
PCFC modules	6	40kW/unit	40×6	240kW	180/900/1200	720–1,080kg	upper center
PCFC efficiency	1	electrical target	-	75% target	-	-	center
AMR liquefier	2	surplus-H2 process	10kg/day×2	process target	240/450/800	250–400kg	F/R
H2 purifier	3	50L/min	50×3	150L/min	180/250/500	120–180kg	F/R
Gas/liquid separator	3	50L/min	50×3	150L/min	200/250/450	120–180kg	F/R
Process-water management	2	condensate / humidity balance	validated setpoint	closed-loop target	150/250/400	80–120kg	front
O2 management membrane	2	optional cathode control	test gate	efficiency TBD	150/250/450	80–120kg	front
CO2 membrane	0	direct-H2 baseline	not required	excluded	-	0kg	rear
Steam separator	2	H2O separation	water balance	condensate recovery	150/250/450	80–120kg	rear
Reformer	0	direct-H2 baseline	not required	excluded	-	0kg	front
Anode loop	1	fuel recirculation	20% target	efficiency gain target	180/250/500	80–120kg	center
LH2 pump	2	50L/min	50×2	100L/min	150/220/350	60–100kg	left
Vaporizer	2	20kg/h	20×2	40kg/h	180/300/600	100–160kg	left
Pressure regulator	4	high→low P	-	4 systems	100/200/300	40–60kg	overall
Shutoff valve	6	safety shutoff	-	6 systems	100/180/250	40–60kg	overall
Vent line	2	safety vent	-	2 systems	100/100/1200	20–40kg	upper
Safety sensors	1	H2/P/T monitor	-	integrated	80/200/300	<30kg	overall
2 Electric/Thermal		PCFC power control	-	battery transient	-	-	center
Inverter	2	120kW	120×2	240kW	150/420/650	180–260kg	upper center
DC/DC	2	100kW	100×2	200kW	130/350/500	120–180kg	upper center
Response battery	1	80kWh	-	80kWh	120/900/1600	350–500kg	upper center
HV J-BOX	1	HV distribution	-	1 system	150/350/500	40–80kg	center
EMS/VCU	1	energy+vehicle ctrl	-	1 system	100/250/300	40–80kg	center
Radiator	2	PCFC/mill/drive	80+19+9	≈108kW	120/400/1200	120–200kg	front
Cooling pump	2	50L/min	50×2	100L/min	120/180/250	40–80kg	overall
HTS cooling line	1	LH2 cooling	-	1 system	60/60/12000	80–120kg	frame
HTC/HTS cable	1	low-loss power	-	1 system	50/50/12000	60–120kg	center→wheel
Waste-heat recovery	1	TE/TPV/Rankine	validated recovery	aux. power TBD	220/500/800	200–350kg	rear
Waste-heat utilization	1	efficiency gain	-	target only	-	-	rear
Power protection	1	insulation/fuse	-	integrated	80/200/300	<30kg	center
3 AD Control		autonomous execution	-	AD operation	-	-	upper
AD computer	1	planning/control compute	validated ODD load	performance TBD by evidence	120/400/500	40–80kg	upper
Sensor suite	1	LiDAR/camera/radar	-	360° sensing	120/500/800	40–80kg	upper
Fusion/localization	1	map+IMU+GNSS	-	position control	80/300/400	<30kg	upper
ODD/DDT monitor	1	state-dependent safety gate	condition class	HOLD / restrict / MRM / STOP	80/250/300	<20kg	center
Dispatch/routing ECU	1	trip execution	-	fleet control	100/300/400	20–40kg	upper
ETA/latency control	1	synchronization	latency≤50ms	AD stability target	80/250/300	<20kg	upper
4 HTS Drivetrain		SC drive+ride ctrl	-	no-sick UX	-	-	corners
LH2 SC-IWM	4	80kW/wheel	80×4	320kW	260/450/450	800–1,200kg	corners
SC regen brake	4	regen braking	-	4 wheels	300/400/400	120–180kg	corners
Air brake	4	braking	-	4 wheels	300/400/400	120–180kg	corners
Knuckle+EPS	4	steer support	-	4 wheels	220/300/400	200–300kg	corners
Air sus.+CFRP	4	vibration cut	-	4 wheels	220/300/400	200–300kg	corners
CTIS/CFRP wheel	4	pressure+light wheel	-	4 wheels	200/450/450	included	corners
Ride comfort target	1	no motion sickness	lateral accel control	UX stabilization	-	-	cabin
5 Structure		flat payload frame	-	380–400mm target	-	-	overall
HTS crossmember	6	beam+cooling	-	6 beams	120/1600/120	200–300kg	lateral
MH main chassis	2	H2 storage+frame	-	≈20kg-H2	300/300/5500	490–860kg	longitudinal
LH2/HTS subframe	2	cooling+support	-	2 systems	200/400/2000	200–400kg	lower
Protection/firewall	1	impact+fire safety	-	integrated	30–80/vehicle	300–600kg	lower/center
Container lock	4	swap fixation	-	4 points	150/300/300	80–120kg	upper
Swap positioning	4	±mm alignment	-	4 points	150/200/200	80–120kg	upper
Service interface	1	refill/drain/inspect	-	water/Ti/slurry	120/400/800	<100kg	L/R
Boarding assist	1	kneel+2-step	−100mm	access assist	200/300/800	120–200kg	right
Payload margin	1	25t GVW−verified curb mass	full integration required	TBD; prior 13–16t claim removed	-	-	overall
Service margin	1	maintenance reserve	total×10–15%	reserve	-	included	overall

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IRR and Cash Flow Structure

This page adds externally sold distributed-HPC and quantum capacity to the reconciled base cash flow. Node plan 4/8/12/20/28/36; HPC CAPEX of ¥5B/¥45B implies ¥8.33M per unit and 150 H100-equivalents/node; paid use rises 35% to 55% at ¥700 to ¥600 per accelerator-hour (1.2kW, PUE 1.15, ¥15/kWh, service 15% of revenue). One quantum module per three-node mesh (1/2/4/6/9/12) costs ¥1.0B (7y) and sells dedicated hours at 20% to 40% paid use and ¥450,000 to ¥375,000/h (USD 3,000 to 2,500 at ¥150), with 25kW, upkeep at 10% of CAPEX and service at 15%. Internal mobility compute is excluded; the case is pre-tax and unlevered. IRR is the discount rate that sets NPV to zero. Project payback is the first year in which cumulative nominal FCF becomes non-negative. Discounted payback is assessed separately. Node payback cannot be calculated because node-level revenue, OPEX, CAPEX, tax, and renewal cash flow are not separately specified.

Investment and Depreciation Structure

Investment / Instrument	Timing	Dep.	Purpose
Initial Development Investment (2025–2030)			
Platform / AI / Urban OS	2025–2030	5y	Core control platform
MaaS Vehicle Prototype	2025–2028	5y	Vehicle feasibility
HTS / Battery / Mill	2026–2030	10y	Energy feasibility
Initial Nodes (4)	2027–2030	15y	Operational deployment
HPC / GTFC	2027–2030	15y	Control infrastructure
Expansion Investment (2030–2035)			
Node expansion (32; 36 total)	2030–2035	15y	City-scale deployment
Fleet Expansion (500)	2030–2035	5y	Utilization scaling
Distributed Energy Infrastructure	2030–2035	10–15y	Hydrogen/electricity scaling
AI / Dispatch Scaling	2030–2035	5y	Real-time optimization
Node Synchronization Infrastructure	2032–2035	15y	Distributed control

Displayed P&L Reference

Year	Revenue	EBITDA	Operating Profit	EBITDA Margin	Operating Margin
2030	¥29.548B	¥12.111B	▲¥8.195B	41.0%	-27.7%
2035	¥133.486B	¥42.553B	¥21.642B	31.9%	16.2%

Annual Cash Flow and Discounted Value

Cash-flow formulas: FCF_t = EBITDA_t − CAPEX_t − ΔWC_t − Cash Taxes_t; DCF_t = FCF_t/(1+r)^t; IRR solves Σ FCF_t/(1+IRR)^t = 0.

Year	Revenue	EBITDA	CAPEX	FCF	Discount factor 8%	Discounted FCF	Cumulative nominal FCF	Cumulative discounted FCF
2025	¥0.000B	▲¥5.000B	▲¥35.000B	▲¥40.000B	1.000	▲¥40.000B	▲¥40.000B	▲¥40.000B
2026	¥0.000B	▲¥6.000B	▲¥40.000B	▲¥46.000B	0.926	▲¥42.593B	▲¥86.000B	▲¥82.593B
2027	¥2.000B	▲¥5.000B	▲¥45.000B	▲¥50.000B	0.857	▲¥42.867B	▲¥136.000B	▲¥125.460B
2028	¥8.000B	▲¥2.000B	▲¥40.000B	▲¥42.000B	0.794	▲¥33.341B	▲¥178.000B	▲¥158.800B
2029	¥18.000B	¥3.000B	▲¥35.000B	▲¥32.000B	0.735	▲¥23.521B	▲¥210.000B	▲¥182.321B
2030	¥29.548B	¥12.111B	▲¥29.000B	▲¥16.889B	0.681	▲¥11.494B	▲¥226.889B	▲¥193.816B
2031	¥47.066B	¥15.595B	▲¥23.000B	▲¥7.405B	0.630	▲¥4.666B	▲¥234.294B	▲¥198.482B
2032	¥66.059B	¥20.219B	▲¥22.000B	▲¥1.781B	0.583	▲¥1.039B	▲¥236.075B	▲¥199.521B
2033	¥87.013B	¥26.455B	▲¥31.200B	▲¥4.745B	0.540	▲¥2.564B	▲¥240.820B	▲¥202.085B
2034	¥110.016B	¥34.309B	▲¥32.200B	¥2.109B	0.500	¥1.055B	▲¥238.711B	▲¥201.030B
2035	¥133.486B	¥42.553B	▲¥32.200B	¥10.353B	0.463	¥4.795B	▲¥228.358B	▲¥196.235B

Financial Results and Return Assessment

Financial result	Value	Interpretation	Decision meaning
Total CAPEX 2025–2035	¥364.600B	Sum of annual CAPEX	Pre-tax / unlevered model
Project IRR	-36.49%	Below 12% hurdle	Return hurdle not met
NPV @ 8%	▲¥196.235B	Discounted value	8% return-hurdle gap
NPV @ 12%	▲¥183.125B	Hurdle-rate value gap	Value uplift required
Nominal project payback	Not achieved	Cum. FCF stays < 0	No renewal CAPEX/tax/WC
Discounted payback	Not achieved	Cumulative DCF remains negative	Separate scenarios required
Node payback	Not calculated	Node-level cash flow absent	Add node economics
2035 EBITDA Margin	31.9%	Displayed P&L	Cash generation before D&A
2035 Operating Margin	16.2%	Displayed P&L	Positive; IRR below hurdle

Sensitivity to Cash Flow Changes

Sensitivity case	Explicit cash-flow change	IRR	NPV @ 8%	NPV @ 12%	Interpretation
No HPC external sales	HPC revenue excluded	-36.39%	▲¥190.321B	▲¥178.298B	Baseline comparison
HPC conservative	100 GPU/node; 40%×¥400/h	-48.22%	▲¥200.365B	▲¥185.983B	Not sufficient
Displayed HPC base	150 GPU/node; 55%×¥600/h	-36.49%	▲¥196.235B	▲¥183.125B	Below no-HPC case
HPC high	200 GPU/node; 70%×¥900/h	-19.93%	▲¥171.651B	▲¥165.064B	Still insufficient
NPV@8% break-even	+¥57.751B/y in 2030–2035	8.00%	¥0.000B	▲¥32.230B	Beyond HPC alone

Nominal Cumulative FCF and Payback Extension

Period	Year / cumulative FCF	Year / cumulative FCF	Year / cumulative FCF
2025–2027	2025 / ▲¥40.000B	2026 / ▲¥86.000B	2027 / ▲¥136.000B
2028–2030	2028 / ▲¥178.000B	2029 / ▲¥210.000B	2030 / ▲¥226.889B
2031–2033	2031 / ▲¥234.294B	2032 / ▲¥236.075B	2033 / ▲¥240.820B
2034–2036	2034 / ▲¥238.711B	2035 / ▲¥228.358B	2036 / ▲¥218.005B
2037–2039	2037 / ▲¥207.652B	2038 / ▲¥197.299B	2039 / ▲¥186.946B
Extension basis	2036–2039	FCF ¥10.353B/year	No renewal CAPEX/tax/WC

Cash-Flow Completeness and Production-Case Inputs

Completeness item	Current treatment	Required production-case addition	Financial effect
Working capital	0	Receivables/payables/inventory timing by year	FCF / payback
Cash tax	0	Loss carry-forward, tax rate and timing	FCF / NPV
Financing	Excluded	Debt/equity, interest, fees and repayment	Equity IRR / DSCR
Residual / salvage	Excluded	Terminal asset value by asset class	Terminal FCF / NPV
Maintenance / renewal	Not separately shown	Overhaul, replacement and lifecycle cadence	OPEX / CAPEX
Node/HPC cash flow	Not separately shown	Revenue/OPEX/CAPEX/tax/LH2 cold energy by node	Node/HPC payback / ROIC

Financial Interpretation & KPI Linkage

The displayed case adds separately priced distributed-HPC and quantum-hour services, excluding internal mobility compute. 2035 HPC/quantum revenue is ¥15.610B/¥15.768B; incremental EBITDA ¥12.730B/¥12.164B. Annual project FCF is positive from 2034, but cumulative nominal FCF remains ▲¥228.358B at the end of 2035. Nominal and discounted payback are not achieved; project IRR is -36.49% and NPV at 8% is ▲¥196.235B (without quantum: -77.97% and ▲¥205.5B). AV-AI training gains from quantum: to verify. The largest unresolved drivers are contracted HPC capacity, paid GPU utilization, accelerator-hour price, node delivery cost, fleet OPEX, hydrogen volume/cost, platform pricing, depreciation-heavy investment, working capital, tax, and renewal CAPEX. KPI improvements affect finance only through explicit volume, price, cost, and investment equations.

Node Deployment Plan

Node Deployment Milestone	Cumulative nodes	Objective
By 2030	4	Initial AVP / MaaS validation
By 2032 (scenario)	12	Fleet synchronization
By 2035	36	City-scale operation

KPI and Financial Equation

KPI	Financial impact	Required equation
Utilization	Revenue increase	available vehicle-hours × monetized trips/hour
Fleet idle ratio	OPEX/CAPEX efficiency	idle hours × avoidable fleet cost
Matching rate	Trips increase	requests × matching × dispatch × realization
Dispatch success	Revenue stability	matched trips × successful dispatch
Energy cost/trip	Margin	energy input × unit cost ÷ trips
Node/HPC utilization	CAPEX efficiency	paid GPU-hours ÷ available external GPU-hours
GPU-hour price	Compute revenue	contracted HPC revenue ÷ paid GPU-hours
Trips/day	Revenue scale	fleet × available hours × trips/hour
Load factor	Revenue efficiency	occupied passenger-km ÷ available seat-km
EBITDA margin	Cash generation	EBITDA ÷ revenue
Operating margin	Accounting profit	operating profit ÷ revenue

Screening-Case Scope Check

Completion Check	Status
P&L / cash-flow asset schedules reconciled	YES (in-service vintages; ¥75.35B CIP at 2030)
H2 internal/external cohorts non-overlapping	YES (internal + external = 100%)
Depreciation rows re-summed	YES
IRR recalculated from displayed FCF	YES
NPV at 8% calculated	YES
Nominal project payback calculated	YES
Discounted payback calculated	YES
Node payback separately calculable	Node-level cash flow required
Rendered Word/PDF layout verified	YES
Working capital included	Excluded from screening case
Cash taxes included	Excluded from screening case
Financing structure included	Excluded from screening case
Residual value included	Excluded from screening case
Renewal CAPEX included	Not separately modeled
Node/HPC revenue/OPEX/CAPEX split	Not separately modeled

Investment Decision Logic

- Autonomous-mobility profitability is dominated by utilization, load, price, energy, fleet OPEX, and infrastructure CAPEX - not labor reduction alone.
- Node deployment may convert parking assets into operational infrastructure only after contract, COGS, utilization, permit, and residual-value assumptions are defined.
- AVP-based infrastructure sharing can support private users and MaaS fleets, but revenue cannot be recognized twice across base fares, platform fees, node fees, and settlements.
- Hydrogen-energy integration can reduce OPEX only if metered end-to-end efficiency, production cost, storage, losses, and uptime support the claimed unit cost.
- Vehicle OEMs, mobility-platform providers, operators..., suppliers, and finance entities require contractually explicit authority and accountability; the responsibility structure is proposed, not current fact.
- At the displayed base, distributed-HPC sales lower project returns versus the no-HPC case and do not achieve nominal payback by 2035; contracted capacity, staged investment, lower CAPEX/OPEX, or alternative financing remains necessary.

Financial Driver Targets

Driver	2030 target	2035 target	Financial impact	Evidence status
Utilization	0.75	0.85	primary volume driver	target
Fleet idle ratio	0.25	0.15	OPEX/CAPEX efficiency	target
Energy cost/trip	≤¥500	≤¥350	margin	target
HPC price/GPUh	¥700	¥600	compute revenue	scenario
Matching rate	≥90%	≥95%	demand realization	target
HPC paid utilization	35%	55%	compute revenue	scenario
Trips/node-day	73	140	revenue scale	scenario
Load factor	0.70	0.80	revenue efficiency	scenario
EBITDA margin	41.0%	31.9%	cash generation	calculated
Operating margin	-27.7%	16.2%	accounting profit	calculated

Investment Assessment

Investment assessment	Target	Displayed case	Status
Nominal project payback	By 2035	Not achieved	BELOW HURDLE
2035 annual FCF	>0	¥10.353B	POSITIVE
IRR	≥12%	-36.49%	BELOW HURDLE
NPV @8%	≥0	▲¥196.235B	BELOW HURDLE
Discounted payback	By 2035	Not achieved	BELOW HURDLE

Distributed-HPC and quantum sales lift 2035 annual FCF to ¥10.353B, but cumulative investment is not recovered by 2035; contracted compute revenue and CAPEX/OPEX optimization remain necessary to meet the 8% NPV and 12% IRR hurdles.

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Design Constraints & System Physics - Source Register

Values are classified as Official value, Technical target, Scenario assumption, or Reference only. A source link does not convert a scenario assumption into a fact. Rows with no page-specific or value-specific support are retained as references but must not be used as validated inputs.

Category	Variable	Corrected value / status	Source	Evidence classification / use
Constraint	Insurance requirement reference	Defined only after exact provision identified	MLIT guidance	Reference only
	Operational-time regulation	Define by service/vehicle/operator law	MLIT transport	Reference only
	Safety-margin requirement	Safety-case derived; no generic numeric value	MLIT safety	Reference only
	Traffic regulation	Applicable rules must be mapped to ODD	NPA self-driving	Official framework
	Weather constraint	ODD-specific threshold required	MLIT AD	Reference only
Cost	Charging infrastructure cost	10–50M JPY/unit retained as scenario range	IEA EV Outlook	Scenario assumption
	EV cost benchmark	5–10M JPY/unit retained as scenario range	IEA EV Outlook	Scenario assumption
	Hydrogen-station CAPEX	150–300M JPY retained as scenario range	IEA Future of H2	Scenario assumption
	Software cost ratio	10–30% retained as scenario range	OECD finance	Scenario assumption; not source-specific
Demand	Reference-market population	Define after the target geography is selected	Statistics Bureau	Official table; value not repeated here
	Japan population	122.68M at 2026-08-01	STAT_POP_202608	Official value
	Taxi fare	Local fare table required; no national 2,000–2,500 constant	MLIT taxi	Official framework
	Trips per person	2.3/day retained only as survey assumption	MLIT transport survey	Scenario assumption
	Target-region vehicle stock	Use latest Chubu table	Chubu vehicle stats	Official table; validate date
Energy	Battery storage cost trend	Declining; exact year/system boundary required	NREL battery	Reference only
	Electricity intraday price	13.0 JPY/kWh retained only with period definition	METI_ELEC_p14	Scenario/period-specific
	Electricity retail price	30.2 JPY/kWh	METI_ELEC_p14	Official cited benchmark
	Electricity spot price	12.24 JPY/kWh retained only with period definition	METI_ELEC_p14	Period-specific
	Hydrogen LHV	33.3 kWh/kg	DOE_H2_LHV	Physical constant
	Hydrogen pathway roundtrip efficiency	Lower than direct electricity; system boundary required	NREL H2 pathway	Reference only
	Hydrogen production cost	150–200 JPY/kg retained as scenario	IEA Future of H2	Scenario assumption
	Hydrogen sales price	300–400 JPY/kg retained as scenario	NEDO H2 library	Scenario assumption
	Hydrogen specific energy	120 MJ/kg	DOE_H2_LHV	Physical constant
	Hydrogen storage cost	DOE 2020 target: $333/kg-H2 stored	DOE_2020_STORAGE	Technical target; not achieved current value
	Hydrogen storage gravimetric	DOE 2020 target: 4.5 wt%	DOE_2020_STORAGE	Technical target; not achieved current value
	PEMFC direct-H2 efficiency	~60% LHV	DOE_PEMFC_60	Technology comparison; not 22%
	LNG energy density	13.9 kWh/kg retained	IEA Future of H2	Reference; page/definition required
	LNG volumetric energy	20.8 MJ/L retained	IEA Future of H2	Reference; page/definition required
	Liquid-H2 volumetric energy	~8 MJ/L retained	DOE H2 storage	Reference
	Generation cost benchmark	LCOE reference only	IEA generation cost	Reference only
	PV high-output benchmark	>1,100kWh/kW/yr	NEDO_PV_p14	Benchmark
	PV output benchmark	1,000–1,100kWh/kW/yr	NEDO_PV_p14	Benchmark
	SOFC efficiency	~60% scenario benchmark	DOE fuel cells	Technology-dependent
Financial	Discount rate	8–12% retained as screening range	OECD finance	Scenario assumption
	Infrastructure return	Project-specific	JBIC	Reference only
	Investment hurdle	Project-specific	OECD finance	Reference only
	Project IRR benchmark	10–15% retained as hurdle range	JBIC	Scenario assumption
Infra	H100 cloud price benchmark	Public cloud ≈$11/GPUh; model ¥400–900/h	Google A3	Benchmark; model is scenario
	GPU/system power	H100 up to 700W; model 1.2kW/GPU + PUE 1.15	NVIDIA H100	Official GPU limit + scenario
	Hydrogen infrastructure	Definition/reference only	IEA Future of H2	Reference only
Logistics	Standard freight fare	Distance/time and vehicle-class tables; no national 52 JPY/kg constant	MLIT_STANDARD_FARE	Official framework
Market	Global EV sales share	15% retained only with year/definition	IEA EV Outlook	Period-specific
	Heavy-truck new sales (Japan)	73k retained only after exact JAMA table validation	JAMA statistics	Validate year/category
	Japan EV sales share	1% retained only with year/definition	IEA EV Outlook	Period-specific
	Total new vehicle sales (Japan)	4.421M retained only after exact JAMA table validation	JAMA statistics	Validate year/category
Policy	Japan 2035 electrified-vehicle target	100% new passenger-car sales; scope definition required	Public policy context	Policy definition required
	Electrified-vehicle definition	BEV/PHEV/HEV/FCEV	METI energy policy	Policy definition
	Japan powertrain distribution	Use a dated JAMA powertrain table	JAMA statistics	Reference only
Regulation	Accident response obligation	Required	NPA_L4	Official
	Administrative cancellation	Applicable	NPA_L4	Official
	Autonomous-driving subsidy availability	Program-specific	MLIT subsidy	Verify FY/eligible costs
	Current safety-policy baseline	Applicable	MLIT safety	Official framework
	Government safety policy	Defined	MLIT AD	Official framework
	Human-operated-vigilance requirement	Depends on operation type	NPA_L4	Official framework
	Infrastructure regulation	Defined by installed equipment/site	MLIT AD	Map exact law
	Insurance requirement	Required; exact coverage required	MLIT AD	Map exact provision
	International standards relevance	Relevant, not automatically legally binding	MLIT safety	Reference
	Legal-liability framework	Defined across product/operator/road law	MLIT safety	Map exact provisions
	Level 4 summary availability	Available	NPA_L4	Official
	Local-government consultation	Required where specified	NPA_L4	Official
	Monitoring requirement	Required where specified	NPA_L4	Official
	Non-specific operation treatment	Defined	NPA interpretation	Official interpretation
	ODD requirement	Required	NPA_L4	Official
	Operation plan submission	Required	NPA portal	Official
	Passenger transport rule	Applicable by service model	e-Gov Passenger Rule	Official law
	Permit cancellation	Applicable	NPA_L4	Official
	Police reporting	Required where specified	NPA_L4	Official
	Public-road operation framework	Defined	MLIT AD	Official
	Remote monitoring	Allowed under defined conditions	NPA_L4	Official
	Road Traffic Act	Applicable	e-Gov Road Traffic Act	Official law
	Road Transport Vehicle Act	Applicable	e-Gov Vehicle Act	Official law
	Road-use regulation	Defined	MLIT safety	Official
	Running-environment condition	Required	MLIT AD	Official
	Specific autonomous-operation permit	Required	NPA portal	Official
	Subsidy eligible cost	Program-specific	MLIT subsidy	Verify current notice
	Suspension authority	Applicable	NPA_L4	Official
	Transport safety law reference	Defined	MLIT safety	Official
	Vehicle Act enforcement order	Applicable	e-Gov order	Official law

Note on Information Sources and Intellectual Property

All content is based on publicly available information and independently constructed models. No confidential or proprietary information from current or previous employers is included. All system architectures, KPIs, financial models, and application prototypes are original and do not rely on internal systems, datasets, or intellectual property.

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